

Machine Learning

Instructor: Hafiz Abdul Rehman

Lecture 08: Performance Measures

Confusion Matrix

$$\text{Accuracy} = \frac{tp + tn}{tp + fp + tn + fn}$$

		Gold Labels	
		Gold Positive	Gold Negative
Predicted Labels	Predicted Positive	True Positives (<i>tp</i>)	False Positives (<i>fp</i>)
	Predicted Negative	False Negatives (<i>fn</i>)	True Negatives (<i>tn</i>)

- You are the CEO of a Pie Company and you need to know what people are saying about your pies on social media
- You build a system that detects tweets concerning your pies
 - the positive class is tweets about your pies
 - the negative class is all other tweets
- Imagine that we looked at a million tweets
 - 100 of them are discussing their love (or hatred) for our pie
 - the other 999,900, are tweets about other matters
- You created a classifier that stupidly classified every tweet as “not about pie”
 - 999,900 true negatives, 100 false negatives, 0 false positives, and 0 true positives, for an accuracy of 999,900/1,000,000 or 99.99%!
- **Accuracy is not a good metric when the goal is to discover a rare event, or at least not completely balanced in frequency**
- **A very common situation in the world!**

Confusion Matrix

		Gold Labels			
		Gold Positive	Gold Negative		
Predicted Labels	Predicted Positive	True Positives (tp)	False Positives (fp)	$\frac{tp}{tp + fp}$	"Precision" aka "Positive Predictive Value"
	Predicted Negative	False Negatives (fn)	True Negatives (tn)	$\frac{tn}{fn + tn}$	"Negative Predictive Value"
		$\frac{tp}{tp + fn}$ "Recall" aka "Sensitivity" aka "True Positive Rate"	$\frac{tn}{fp + tn}$ "Specificity" aka "True Negative Rate"	$Accuracy = \frac{tp + tn}{tp + fp + tn + fn}$	

Confusion Matrix

		Gold Labels			
		Gold Positive	Gold Negative		
Predicted Labels	Predicted Positive	True Positives (tp)	False Positives (fp)	$\frac{tp}{tp + fp}$	"Precision" aka "Positive Predictive Value"
	Predicted Negative	False Negatives (fn)	True Negatives (tn)	$\frac{tn}{fn + tn}$	"Negative Predictive Value"
		$\frac{tp}{tp + fn}$ "Recall" aka "Sensitivity" aka "True Positive Rate"	$\frac{tn}{fp + tn}$ "Specificity" aka "True Negative Rate"	$Accuracy = \frac{tp + tn}{tp + fp + tn + fn}$	
		$\frac{fn}{tp + fn}$ 1 - Sensitivity = "False Negative Rate" aka "False Rejection Rate"	$\frac{fp}{fp + tn}$ 1 - Specificity = "False Positive Rate" aka "False Acceptance Rate"	$\frac{\text{True Positive Rate}}{\text{False Positive Rate}} = \text{"Positive Likelihood Ratio"}$ $\frac{\text{False Negative Rate}}{\text{True Negative Rate}} = \text{"Negative Likelihood Ratio"}$	

Confusion Matrix

- Sensitivity, specificity, FNR and FPR are not influenced by real-world data imbalances
 - These imbalances impact the denominators
 - E.g. a rare disease, or a rare phenomena (like a fraud email)

	Actual +	Actual -
Test +	5	5,000
Test -	5	5,000

$$Sensitivity = \frac{5}{10} = 0.5, Precision = \frac{5}{5 + 5000} = 0.0009$$

- Precision and Negative Predictive Value are impacted by these imbalances and are sensitive to them.

The Thresholding Problem in Classification

- Pinocchio's nose
 - Assume that liars have longer noses (wooden dummies only! 😊)
 - We need to set a threshold on the nose length (input feature) above which we classify the subject as a liar (label = yes).
 - The same principle applies to classification of:
 - A growth as a tumor based on size
 - Blood pressure levels as hypertension
 - An email as SPAM/Not-SPAM, Misinfo/Not-Misinfo based on probability scores
 - A search item as match/not-match based on similarity scores, e.g.
 - spoken term detection
 - keyword spotting
 - voice biometrics
- So, the question is where to place the cutoff
 - Can we exhaustively try all cutoffs over a bounded score?
 - Yes, but what do we track?
 - Precision/Recall?
 - False acceptances/False rejections?
 - True positives/False positives?
- Say hello to RoC curves!

Receiver Operating Characteristic (RoC)

- A graphical plot that illustrates the diagnostic ability of a binary classifier as its discrimination threshold is varied
- The method was originally developed for operators of military radar receivers starting in 1941, which led to its name.
- Plot the **true positive rate (TPR)** – sensitivity – against the **false positive rate (FPR)** – (1 - specificity) at various threshold settings

Receiver Operating Characteristic (RoC)

Height h (inches)	Output Score (probability)	Adult (gold)
12	0.12	n
82	0.82	y
18	0.18	n
60	0.6	y
72	0.72	y
55	0.55	n
48	0.48	y
24	0.24	n
26	0.26	n
68	0.68	y

Receiver Operating Characteristic (RoC)

Height h (inches)	Output Score (probability)	Adult (gold)	Thresholds																			
			0.1		0.2		0.3		0.4		0.5		0.6		0.7		0.8		0.9		1.0	
12	0.12	n																				
82	0.82	y																				
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82	0.82	y	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	n	fn	n	fn
18	0.18	n	y	fp	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn
60	0.6	y	y	tp	y	tp	y	tp	y	tp	y	tp	n	fn	n	fn	n	fn	n	fn	n	fn
72	0.72	y	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	n	fn	n	fn	n	fn
55	0.55	n	y	fp	y	fp	y	fp	y	fp	y	fp	n	tn	n	tn	n	tn	n	tn	n	tn
48	0.48	y	y	tp	y	tp	y	tp	y	tp	n	fn	n	fn	n	fn	n	fn	n	fn	n	fn
24	0.24	n	y	fp	y	fp	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn
26	0.26	n	y	fp	y	fp	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn	n	tn
68	0.68	y	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	n	fn	n	fn	n	fn	n	fn

Receiver Operating Characteristic (RoC)

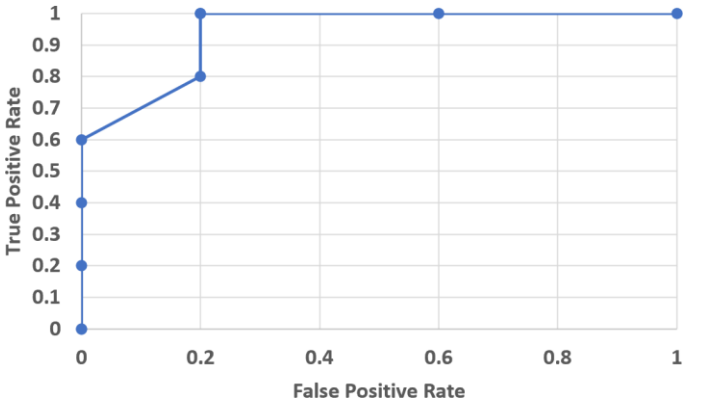
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68	0.68	y	y	tp	y	tp	y	tp	y	tp	y	tp	y	tp	n	fn	n	fn	n	fn	n	fn
tp			5		5		5		5		4		3		2		1		0		0	
fn			0		0		0		0		1		2		3		4		5		5	
fp			5		3		1		1		1		0		0		0		0		0	
tn			0		2		4		4		4		5		5		5		5		5	

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tp			5		5		5		5		4		3		2		1		0		0	
fn			0		0		0		0		1		2		3		4		5		5	
fp			5		3		1		1		1		0		0		0		0		0	
tn			0		2		4		4		4		5		5		5		5		5	
1-Specificity (FPR)			1		0.6		0.2		0.2		0.2		0		0		0		0		0	
Sensitivity (Recall, TPR)			1		1		1		1		0.8		0.6		0.4		0.2		0		0	
Precision			0.5		0.625		0.8333		0.8333		0.8		1		1		1		NAN		NAN	

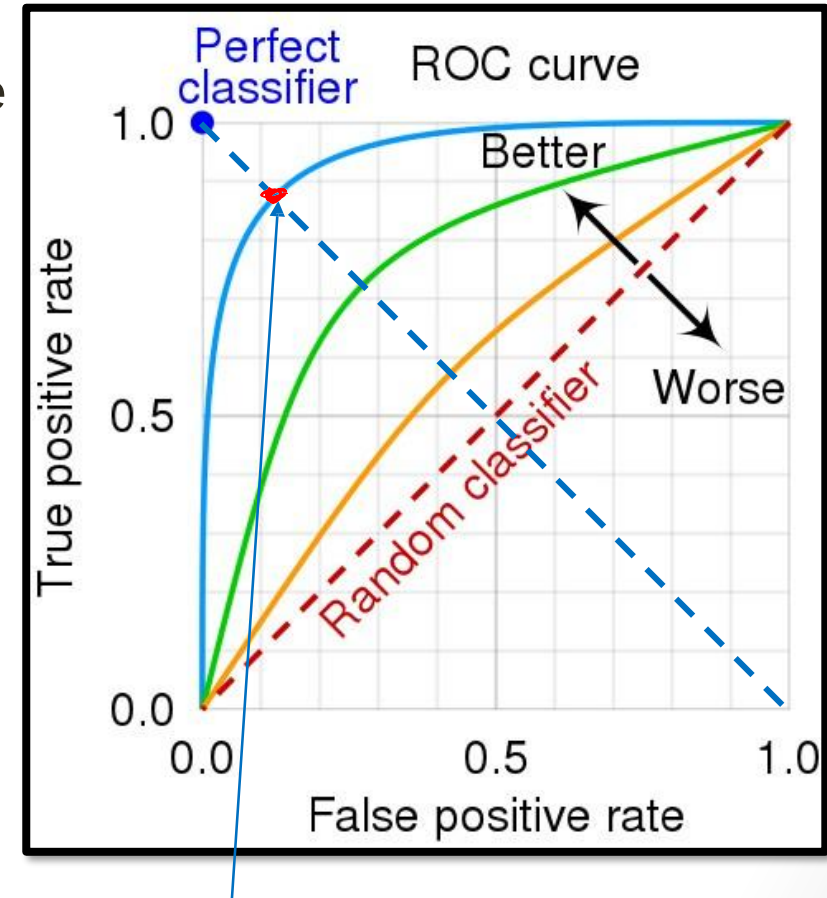
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tp			5		5		5		5		4		3		2		1		0		0	
fn			0		0		0		0		1		2		3		4		5		5	
fp			5		3		1		1		1		0		0		0		0		0	
tn			0		2		4		4		4		5		5		5		5		5	
1-Specificity (FPR)			1		0.6		0.2		0.2		0.2		0		0		0		0		0	
Sensitivity (Recall, TPR)			1		1		1		1		0.8		0.6		0.4		0.2		0		0	
Precision			0.5		0.625		0.8333		0.8333		0.8		1		1		1		NAN		NAN	



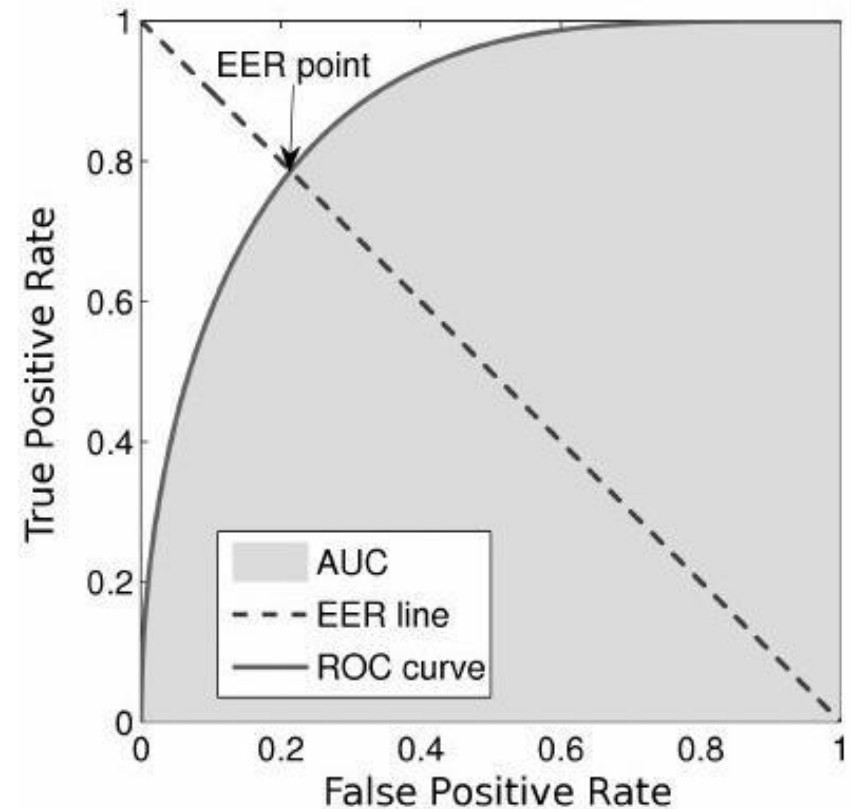
Characteristics

- The best possible prediction method would yield a point in the upper left corner (0,1) i.e. 100% sensitivity (no false negatives) and 100% specificity (no false positives)
- A random guess would give a point along a diagonal line (the line of no-discrimination) from the left bottom to the top right corners (TPR = FPR)
- The **red diagonal** divides the ROC space.
 - Points above the diagonal represent good classification (better than random)
 - Points below the line represent bad results (worse than random)
 - The output of a consistently bad predictor could simply be inverted to obtain a good predictor.
- The **blue diagonal** is the Equal Error Diagonal
 - Here $FPR = FNR$
 - Where $FNR = 1 - TPR$
- A viable way to locate desired threshold: Smaller is better (in the graph: higher and to the left)

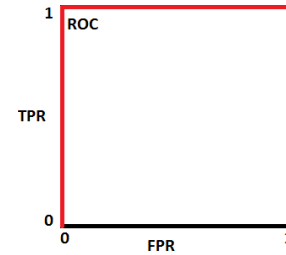
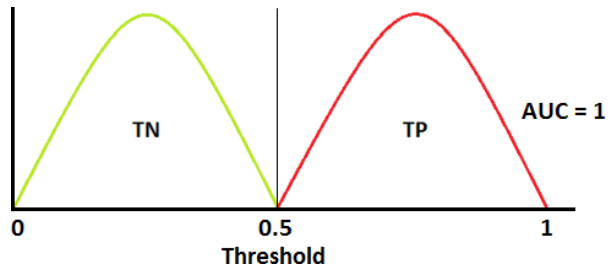


More Characteristics

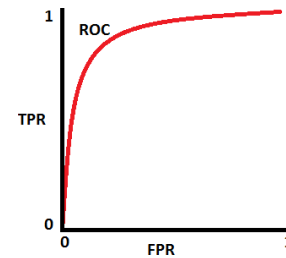
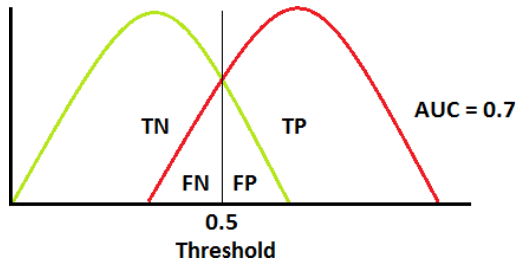
- It is hard to compare classifiers using ROC curves
- A way around that is to use AUC – Area under the ROC Curve, A' (pronounced "a-prime") or "c- statistic" ("concordance statistic").
 - Larger is better
- “AUC ROC can be interpreted as the probability that the scores given by a classifier will rank a randomly chosen positive instance higher than a randomly chosen negative one.” (Page 54, Learning from Imbalanced Data Sets, 2018)
- For imbalanced datasets: “ROC analysis does not have any bias toward models that perform well on the minority class at the expense of the majority class—a property that is quite attractive when dealing with imbalanced data.” (Page 27, Imbalanced Learning: Foundations, Algorithms, and Applications, 2013)



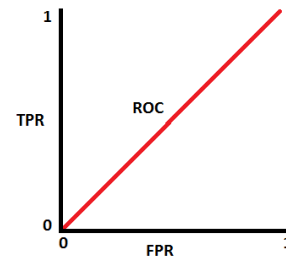
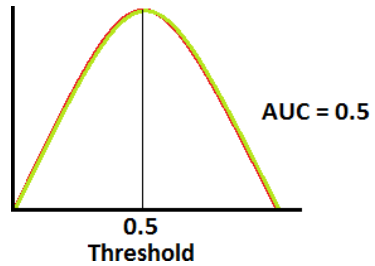
Examples



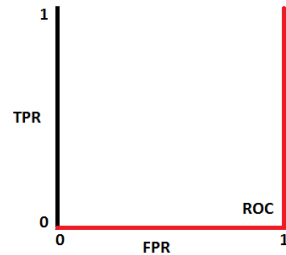
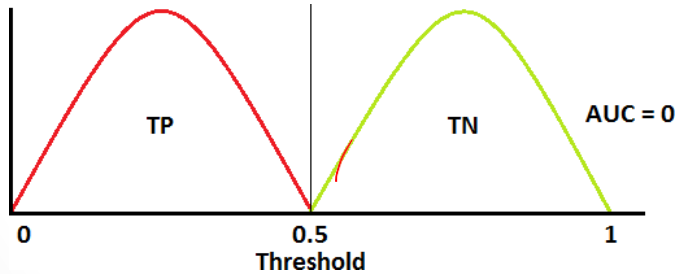
$$AUC \rightarrow 1$$



$$AUC \approx 0.7$$



$$AUC \approx 0.5$$

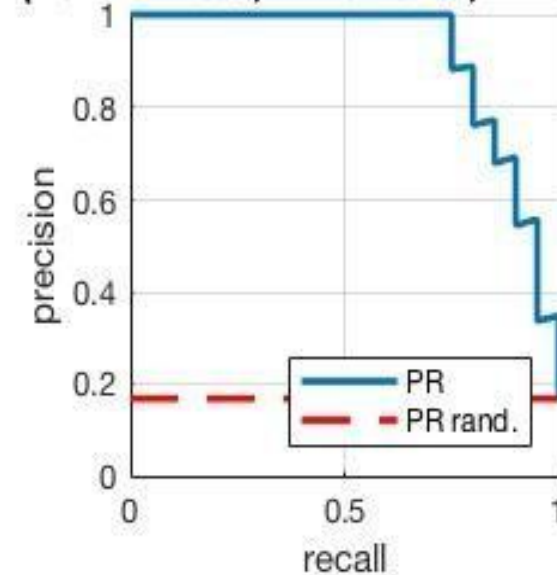


$$AUC \rightarrow 0$$

Precision-Recall Curve

- The precision-recall curve shows the tradeoff between precision and recall for different thresholds. A high area under the curve represents both high recall and high precision
- High scores for both show that the classifier is returning accurate results (high precision), as well as returning a majority of all positive results (high recall).

PR (AUC: 91.18%, AP: 91.32%, AP11: 90.29%)



Discussion: PR vs ROC

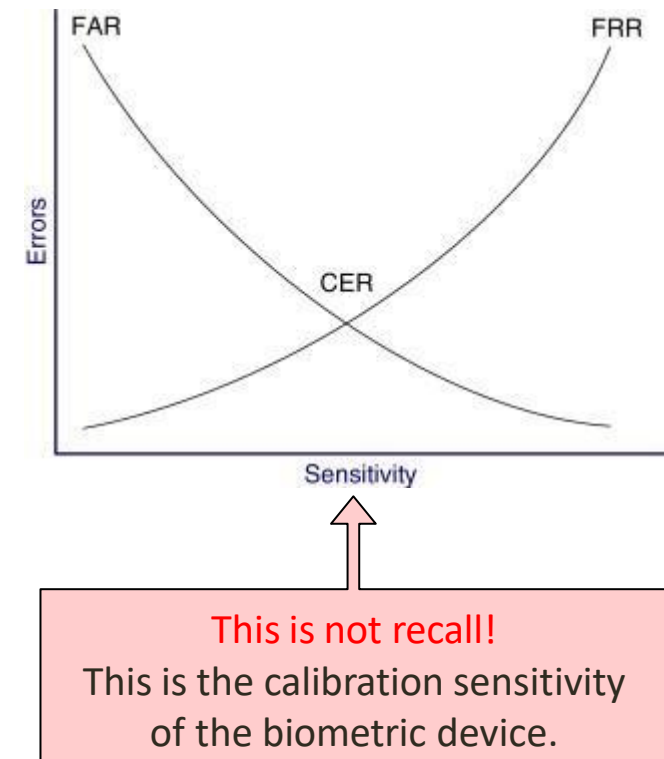
- Assuming a "positive" class 1 and a "negative" class 0. $\hat{y}$ is our estimate of the true class label y .
- $P(y = 1)$ is the baseline probability depending on how common the event is in the real world
- $P(\hat{y}=1)$ is the probability that our classifier will classify a sample as positive
 - **Precision** = $P(y = 1 | \hat{y} = 1)$
 - **Recall = Sensitivity** = $P(\hat{y} = 1 | y = 1)$
 - **Specificity** = $P(\hat{y} = 0 | y = 0)$
- ROC curves will be the same regardless of $P(y = 1)$
- PR curves may be more useful in practice for needle-in-haystack type problems or problems
- where the "positive" class is more interesting than the negative class.

Bottomline:

- Use precision and recall to focus on a small positive class
 - When $P(y = 1)$ is low and the ability to detect correctly positive samples is our main focus (correct detection of negatives examples is less important to the problem).
- Use ROC when both the detection of both classes is equally important
 - When we want to give equal weight to the prediction ability of both classes.
- Use ROC when the positives are the majority or switch the labels and use precision and recall
 - When the positive class is larger, use the ROC metrics because the precision and recall would reflect mostly the ability of prediction of the positive class and not the negative class which will naturally be harder to detect due to the smaller number of samples.
 - If the negative class (the minority in this case) is more important, we can switch the labels and use precision and recall.

A word on Biometric Systems Measure

- The crossover error rate describes the point where the false reject rate (FRR) and false accept rate (FAR) are equal. CER is also known as the equal error rate (EER). The crossover error rate describes the overall accuracy of a biometric system.
- As the **sensitivity** of a biometric system increases, FRRs will rise and FARs will drop. Conversely, as the sensitivity is lowered, FRRs will drop and FARs will rise.
- CER is better when lower.
- Authentication algorithms need to simultaneously minimize the *permeability* to intruders, therefore they have to be *demanding*, and to maximize the *comfort* level, therefore to be *permissive*.
 - This contradiction is the base for the optimization problem in authentication algorithms and the measure of success for the overall precision of an algorithm and of its usability is the CER.



Thank You